

Swiss Institute of Bioinformatics

SINGLE-CELL TRANSCRIPTOMICS WITH R

Group work & using agents

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Group Work

- **Work in groups to analyse a scRNAseq dataset**
  4 different projects

- **Go through the project questions** (use the course materials)



Group Work

Group work: everyday at set times (flexible)

- **Last day: present your work**

Prepare a few slides:

- Main methods and results (include visualisations)
- Challenges (QC, methods, interpretations..)
- Findings/conclusions



Four different projects

Project 1 - Green Fluorescent Zebrafish

Project 2 - *Drosophila* on Cocaine

Project 3 - PBMCs of melanoma patients

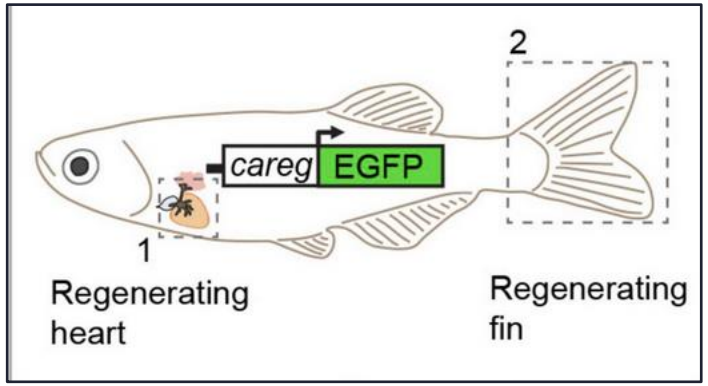
Project 4 - Cervical cancer and HPV

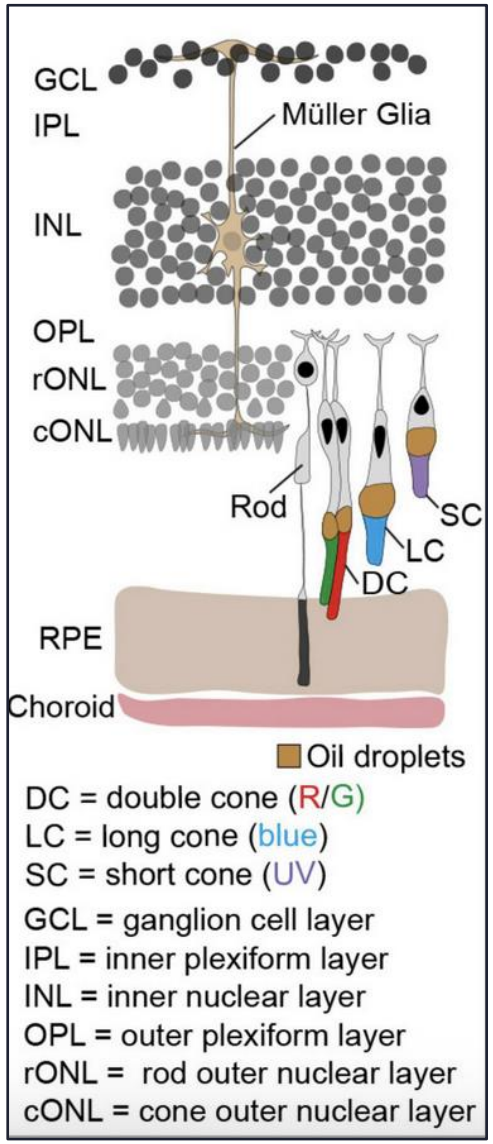
Project 1: Zebrafish

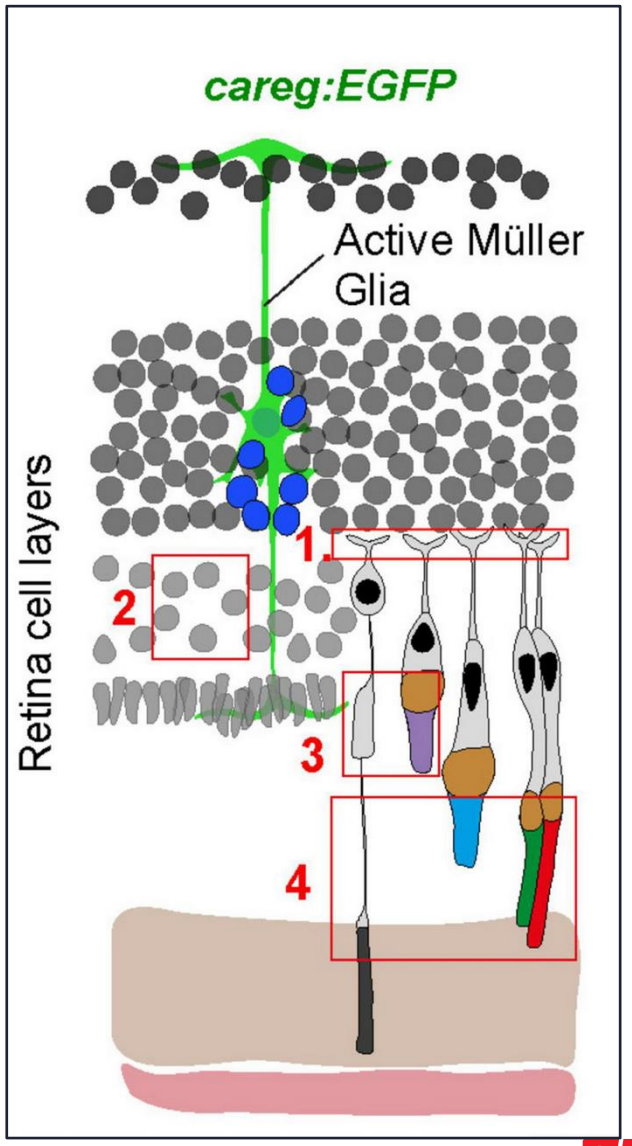
The regeneration-responsive element *careg* monitors activation of Müller glia after MNU-induced damage of photoreceptors in the zebrafish retina

Thomas Bise et al. 2023

Careg reporter detects activated MG, and provides a common marker of regeneration-competent cells





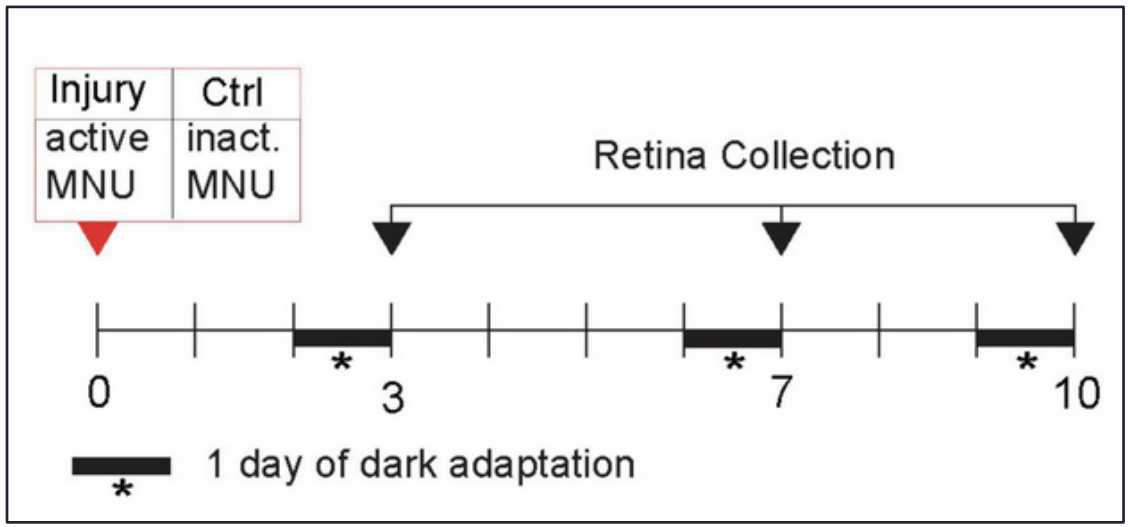


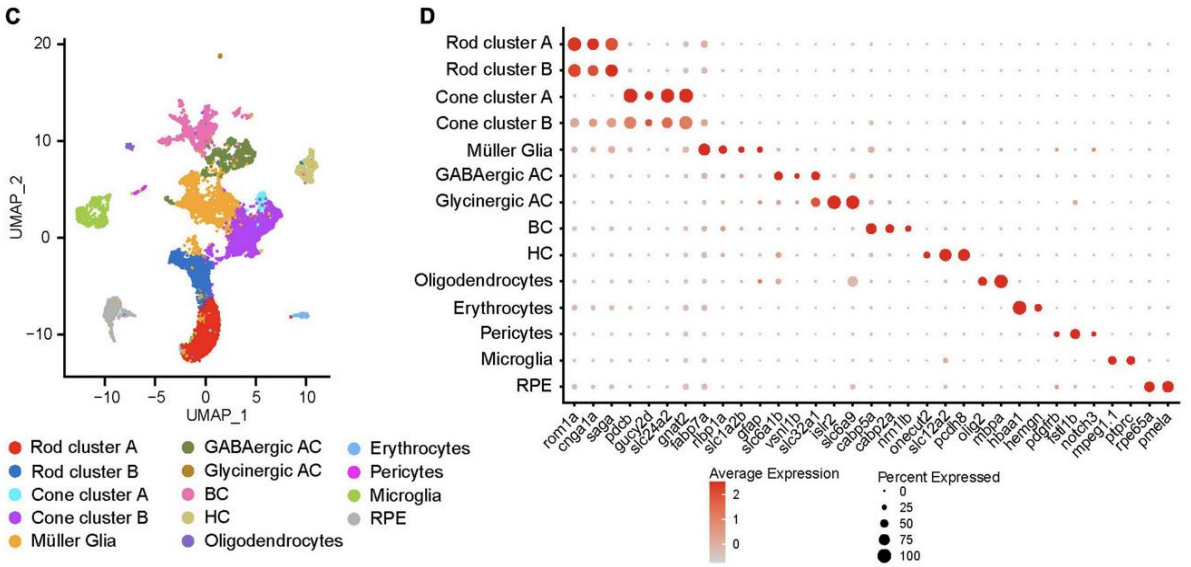
Project 1: Zebrafish

Species: *Danio Rerio*

Tissue: Retina

Conditions: Control, damaged tissue (3,7,10 days)





- Can you annotate the main cell types of interest: MG, cons and rods?
- Identify activated MG in the different conditions
- Look at molecular differences between injured and uninjured cons or rods

Project 2: *Drosophila* on cocaine

Study Focus: Investigating **single-cell transcriptional responses** in the *Drosophila* brain after acute cocaine exposure.

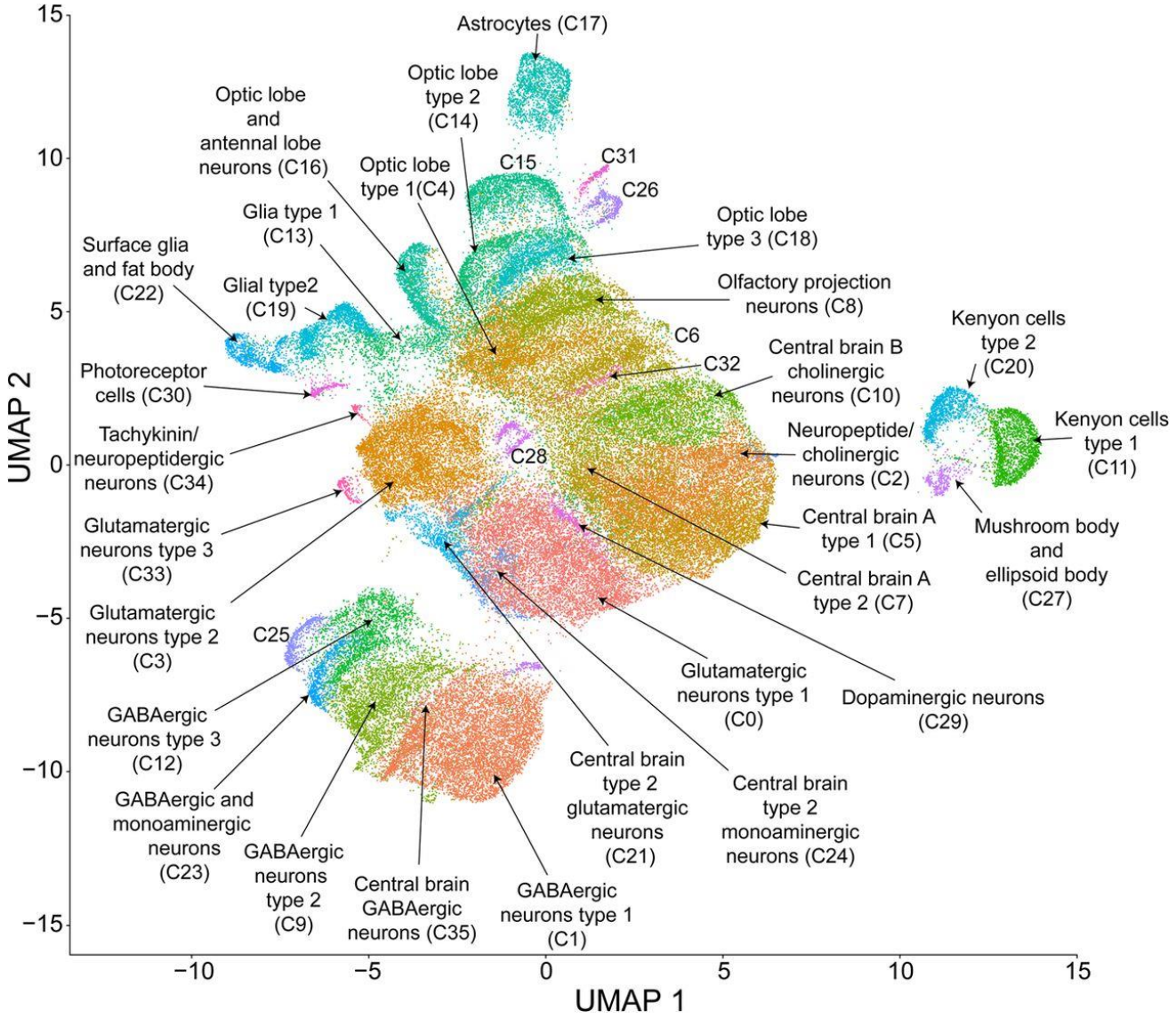
Unsupervised clustering of 86,224 cells revealed **36 distinct clusters**, representing major neuronal and glial cell types across brain regions.

Profound Sexual Dimorphism: Males exhibited more pronounced and widespread transcriptional changes to cocaine than females.

Available Data: Pre-processed single-cell count matrices (GEO GSE152495) for male/female, cocaine/sucrose conditions, ready for analysis.



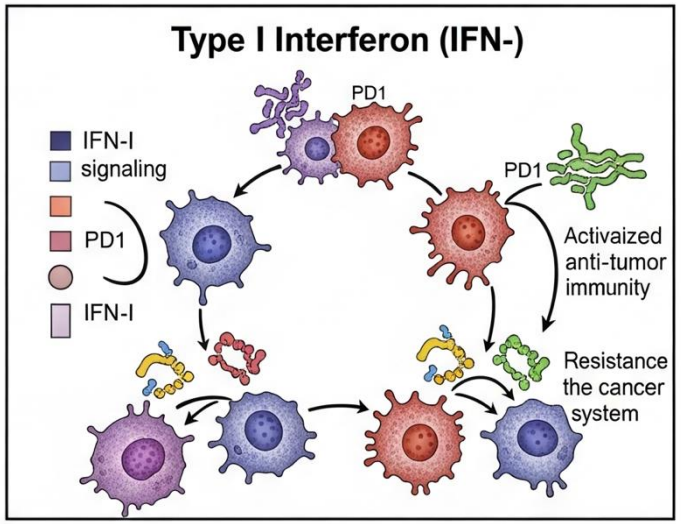
P2: UMAP visualization and clustering



This figure will be very useful when you are performing the "Dimensionality Reduction" and "Unsupervised Clustering" steps, and especially for "Cluster Annotation" to compare your results and ensure you are identifying similar cell populations and their spatial relationships in the UMAP embedding.

Brandon M. Baker et al. Genome Res. 2021;31:1927-1937

Project 3 - IFN-I Responsiveness & PD1 Blockade



"PD1 Blockade" refers to a specific type of cancer immunotherapy that aims to reactivate immune cells to fight cancer.

IFN-I are alarm signals that cells release when they detect threats like viruses or certain aspects of cancer.

Question: Why some cancer patients respond well to a type of immunotherapy called PD1 blockade, while others don't?

The Problem: Immunotherapy like PD1 blockade can be very effective against cancer, but it doesn't work for everyone.

Project 3 - IFN-I Responsiveness & PD1 Blockade

Our Approach: Analyzing single-cell RNA sequencing (scRNA-seq) data from healthy donors and 8 treated patients.

Key Finding: Patients with lower pre-therapy IFN-I responsiveness in CD4/CD8 effector T cells (Teff) showed:

- Signatures of improved immune function.
- Better therapy outcomes.

Available Data: Pre-processed scRNA-seq count matrices from GEO GSE199994.

Project 4 – Cervical Cancer Dataset

Aim 1 (paper 1): Understand the mechanisms of cervical cancer development mediated by HPV infection

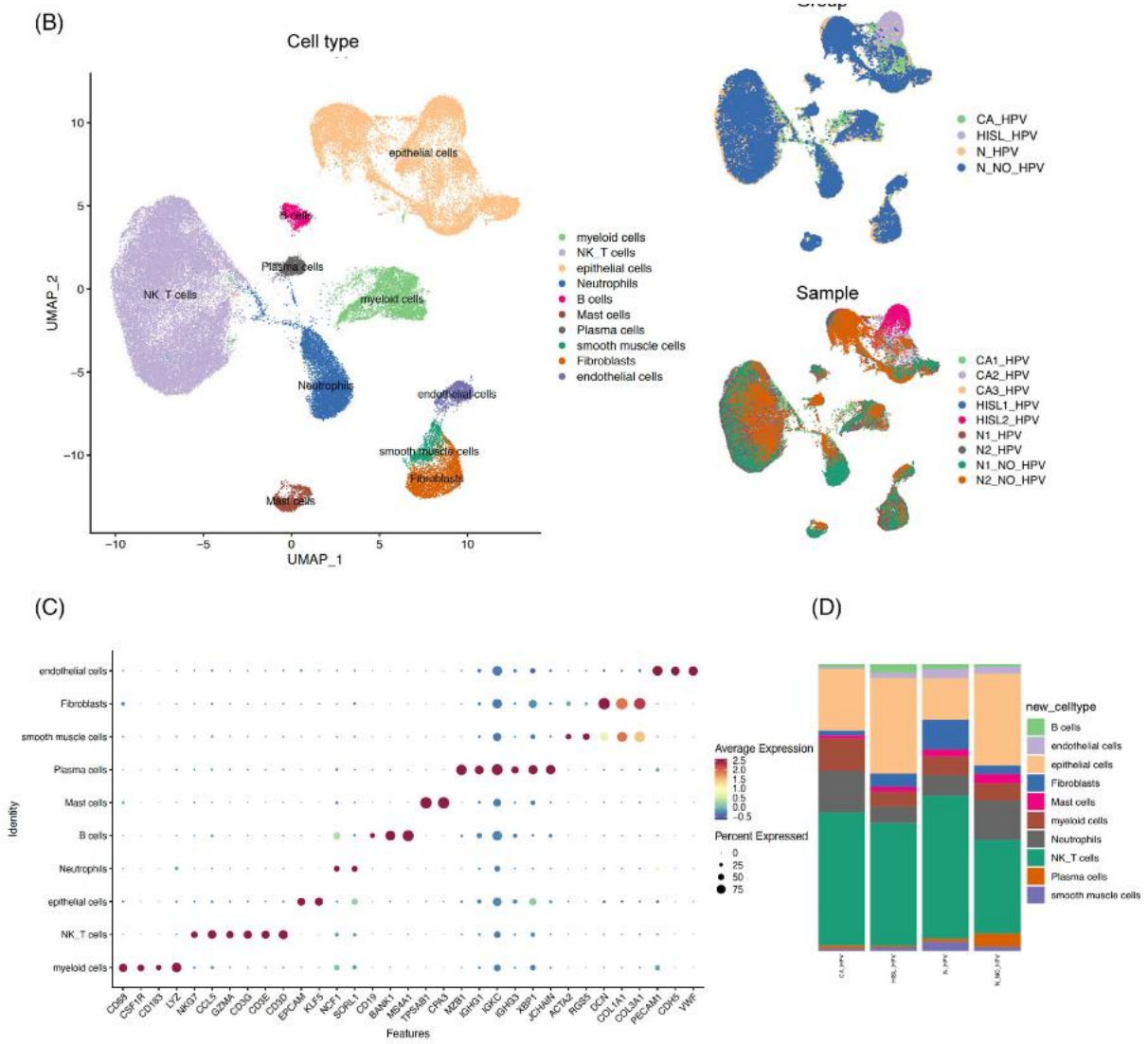
Aim 2 (paper 2): Understand the mechanisms of immune remodelling in cervical cancer

Data:

- Normal cervix (no HPV) (n = 2)
- Cervical cancer (with HPV) (n = 2)
- Normal cervix (with HPV) (n=2)
- Pre-cancer cervix (with HPV) (n=2)

} Use these samples only

P4: Key Figure (try to reproduce)



Source: Guo (2023) - *Paper 1*

Choose a project: forms.office.com/e/2KyGnwnv28

Project 1 - Green Fluorescent Zebrafish

Project 2 - Drosophila on Cocaine

Project 3 - PBMCs of melanoma patients

Project 4 - Cervical cancer and HPV

You may use agentic AI for the group work

- **LLM**: a complex model that takes a specific type of **input** (usually text or images) and writes a specific type of **output** (usually text)
- **Harness**: (also often referred to as agent) software that can **perform tasks** on your computer based on the context, input of the user and output of an LLM
- (**Coding**) **agent**: LLM + Harness

- We have prepared coding agents to use in your group work
- Do **not** use a coding agent during the regular exercises!
- More importantly: it comes with an extensive set of packs specifically for this course that form instructions to the agent: LLM packs

Questions

The setup

- LLM packs: for each step in the analysis, there is an extensive set of instructions that can be used by any agent. These help you doing a trustworthy analysis – however, you are still responsible.
- We have set up Posit Assistant (within Rstudio) and Goose (from the command line) for the course. They use Deepseek 4.1 at openrouter ($0.006/1M tokens).
- Install the LLM packs by giving your agent instructions:

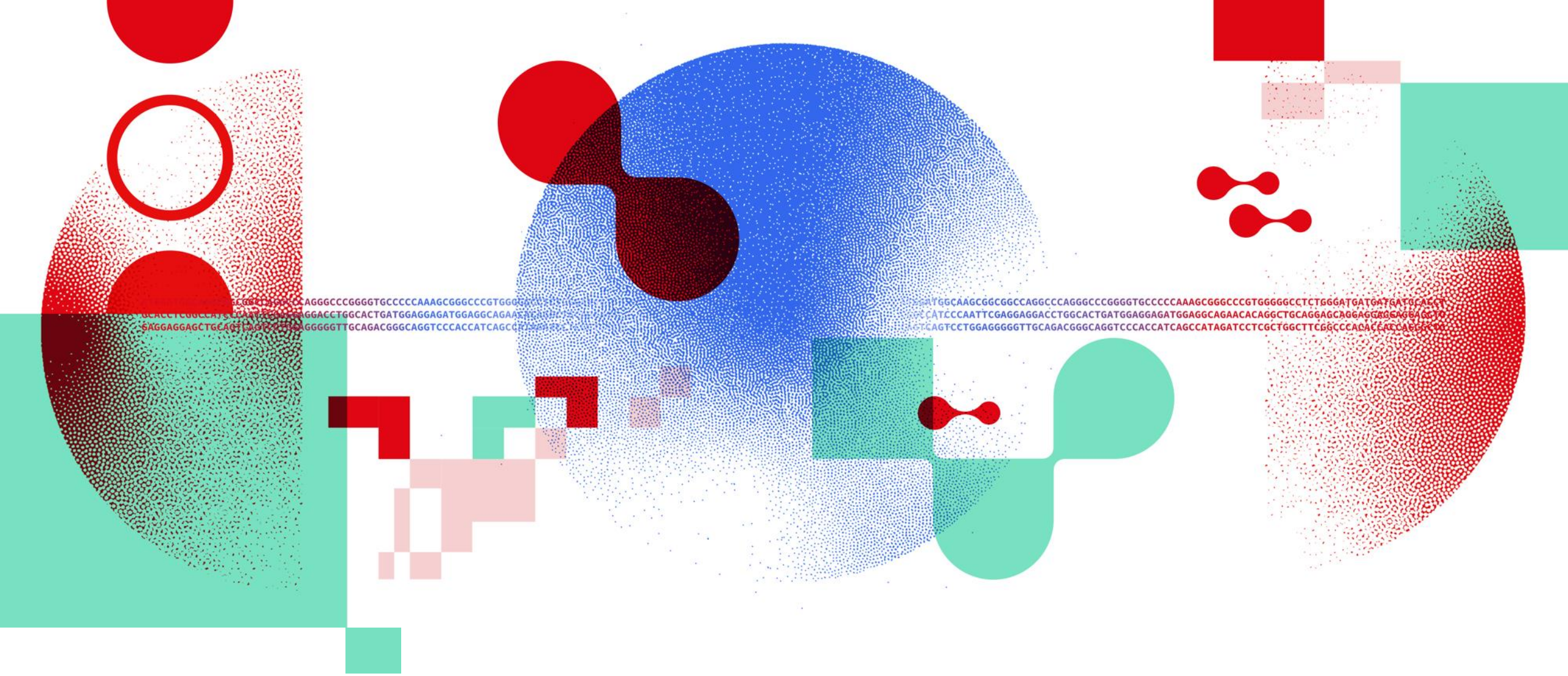
Set up all of the LLM packs from the SIB single-cell transcriptomics course (https://github.com/sib-swiss/single-cell-r-training) in this project, without downloading the whole course repository:

1. In a temporary directory, sparse-clone only the llm-packs folder (not llm-packs/_dev, which is authoring material, not a pack), using `git clone --filter=blob:none --sparse` followed by `git sparse-checkout set llm-packs` and then excluding llm-packs/_dev.

....

Why an LLM pack?

- Providing instructions to an agent to do an analysis from scratch can easily lead to:
  - Hallucinations
  - Misinterpretations
  - Code/output that you do not understand
- The LLM packs we provide are specifically tailored **to this course**; it should generate output that **you understand** and can **verify**, and ask for **decisions** that you should be able to make
- Using a specific set of instructions can tremendously **reduce** the number of **tokens** used



Thank you

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